EX: 14

10/10/26

Write a code using RAW Sockets TO Implement Packet Sniffing

AIM:

To implement a packet sniffing program using RAW sockets to capture and display network packets.

ALGORITHM:

* Start the program.
* Create a RAW socket.
* Receive network packets continously.
* Extract the Ethernet header.
* Identify the packet protocol
* Extract source and destination IP address for IPV4 packets.
* Display packet details
* Stop the program with Ctrl+C
* Close the socket.

Program:

```
import socket
import struct
raw_socket = socket.socket(
    socket.AF_PACKET,
    socket.SOCK_RAW,
    socket.ntohs(3)
)
print("Packet Sniffer Started...")
```

```
while True:
    packet, address = raw_socket.recvfrom(65535)
    dest_mac, src_mac, protocol = struct.unpack(
        "!6s6sH", packet[:14]
    )
    dest_mac = ":".join("%.02x" %b for b in dest_mac)
    src_mac = ":".join("%.02x" %b for b in src_mac)
    print("=========")
    print("Source MAC :", src_mac)
    print("Destination MAC:", dest_mac)
    print("protocol :", hex(protocol))

    if protocol == 0x0800:
        ip_header = packet[14:34]
        fields = struct.unpack(
            "!BBHHHBBH4s4s", ip_header
        )
        src_ip = socket.inet_ntoa(fields[8])
        dest_ip = socket.inet_ntoa(fields[9])
        print("Source IP :", src_ip)
        print("Destination IP :", dest_ip)
        print("TTL :", fields[5])
        print("IP protocol :", fields[6])

    elif protocol == 0x0806:
        print("packet TYPE : ARP")
```

OUTPUT:

Packet Sniffer Started....
waiting for packets....

Source MAC : 3C:52:82:11:22:33
Destination MAC : ff:ff:ff:ff:ff:ff
protocol :0x806
packet TYPE : ARP

Source MAC : 3C:52:82:11:22:33
Destination MAC : 84:16:f9:44:55:66
protocol :0x800
Source IP : 192.168.1.10
Destination IP : 192.168.1.1
TTL : 64
Ip protocol : 6

Packet Sniffing Stopped.

```
elif protocol == 0x86DD:
    print("Packet Type : IPv6")
else:
    print("Packet Type : Other")
except KeyboardInterrupt:
    print("\n packet Sniffing Stopped:")
finally:
    raw_socket.close()
```

RESULT:

Thus, the packet sniffing program was successfully implemented using RAW sockets to capture network packets and display their MAC addresses, IP addresses, protocol types

Bill 14/4/24